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Block Course Cosmology

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Problem Sheet #2 (Hand out: Fri => hand in: Mon)

1. Build your own Python Cosmological Calculator (Part I)

1.1 Assume an Einstein-deSitter Universe ($\Omega_m = 1$, $\Omega_{\Lambda} = 0$, i.e. flat with $k=0$), and write Python functions that return

- The Hubble Factor
- The scale factor
- The Density parameters $\Omega_i = \rho_i/\rho_{\text{crit}}$
- The proper i.e. physical and comoving distances
- The cosmic time

as a function of redshift.

Plot $H(z)$, $\Omega_i(z)$, $d_{\text{Com}}(z)$, $d_{\text{Phys}}(z)$, $t(z)$ vs. redshift z and $a(t)$ vs. t in different panels in the case of the Einstein-deSitter Universe. Confirm the behaviors and findings discussed in the lectures.

1.2 What can you say about the evolution of a matter-dominated Universe for different values of Ω_m ?

Namely, generalize the functions above, so that it accepts as arguments not only redshift but also Ω_m . Then plot $H(z)$, $\Omega_X(z)$, $d_{\text{Com}}(z)$, $d_{\text{Phys}}(z)$, $t(z)$ for a handful of different cosmological scenarios: e.g. $\Omega_m = 1$, $\Omega_m = 0.7$, $\Omega_m = 0.3$.

1.3 Generalize the above functions to any non-flat cosmological model with a cosmological constant Λ , i.e. to any combinations of the cosmological parameters Ω_m , Ω_r , Ω_{Λ} , h , ...

Compare the functional forms of $H(z)$, $\Omega_X(z)$, $d_{\text{Com}}(z)$, $d_{\text{Phys}}(z)$, $t(z)$ vs. redshift z and $a(t)$ vs. t for an Einstein-deSitter Universe ($\Omega_m = 1$, $\Omega_{\Lambda} = 0$) and a concordance model Λ CDM (aka standard cosmological model Λ CDM).

Note: the currently accepted values for the cosmological parameters in our Universe are in the table: the current Universe is compatible with a flat Λ CDM model. Please use these values with you solve your functions numerically and want actual values. The density parameter of the radiation can be taken to be:

$$\Omega_{\text{rad}} = 2.47 \times 10^{-5} h^{-2}.$$

Table 8.2 Summary of the cosmological parameters as determined by the Planck 2013 data, in combination with low- ℓ polarization measurements from WMAP (WP)

Ω_{Λ}	0.685 ± 0.018
Ω_m	0.315 ± 0.018
$\Omega_b h^2$	0.02205 ± 0.00028
$H_0 (\text{km s}^{-1} \text{Mpc}^{-1})$	67.3 ± 1.2
z_{ion}	11.1 ± 1.1
τ	0.089 ± 0.014
σ_8	0.829 ± 0.012
Y_p	0.24770 ± 0.00012
$t_0 (\text{Gyr})$	13.817 ± 0.048
z_{rec}	1090 ± 0.54
z_{eq}	3391 ± 60

2. The evolution of the densities of the various components of the Universe.

2.1 How do the densities of matter, radiation, and dark energy (i.e. of a cosmological constant, for now) evolve with redshift in the following Cosmological scenarios?

- Einstein - DeSitter Universe ($\Omega_m = 1$, $\Omega_{\Lambda} = 0$)
- Low-density Universe ($\Omega_m = 0.3$, $\Omega_{\Lambda} = 0$)
- Standard Cosmological Universe ($\Omega_m = 0.3$, $\Omega_{\Lambda} = 0.7$, $h=0.7$ or Table)

Express everything in units of the critical density and make plots of the three density parameters as a function of redshift in the three aforementioned cosmological scenarios. NOTE: when not specified, by densities of matter, radiation, and dark energy, we mean proper i.e. physical densities (i.e. not comoving).

2.2 In the standard cosmological scenario, what is the redshift of the three equivalence epochs? (The equivalence between energy component i and j occurs when $\rho_i = \rho_j$, where i = matter, radiation, cosmological constant). Identify the radiation, matter and dark-energy dominated eras: what redshift ranges do they correspond to? At what redshift was the Sun born? At what redshift did the dinosaurs disappear?

Solve this by interpolating the required function as a function of time/redshift.

3. The age of the Universe and the standard cosmological scenario

3.1 Our universe is consistent with : $\Omega_m \sim 0.3$; $\Omega_{\Lambda} \sim 0.7$; $h \sim 0.7$. as of the cosmological parameters today being

By using the values in the inset or those of the table above, please answer the following.

Determine:

- The age of the Universe.
- The age at which the expansion of the Universe first started to accelerate.
- The age at which the energy density in matter and the cosmological constant were equal.

3.2 A cosmologically-ignorant civilization measures $h=0.7$ but refuses to countenance the possibility of a cosmological constant. How old would they believe their Universe to be if they assumed it was flat with $\Omega_m = 1$?

4. Galaxies, redshift, and cosmic time

Galaxies are observed nowadays all the way up to $z=13$. How old was the Universe at the time from which we see such most distant galaxies? What fraction of the universe history does that correspond to? How far in comoving and physical distance are those galaxies from us?

NOTE: the tools developed to solve this Problem Sheet will be needed for Problem Sheets #3, #4, and #5.

NOTE: you may not use Astropy or similar cosmology-related existing Python libraries to solve these exercises.